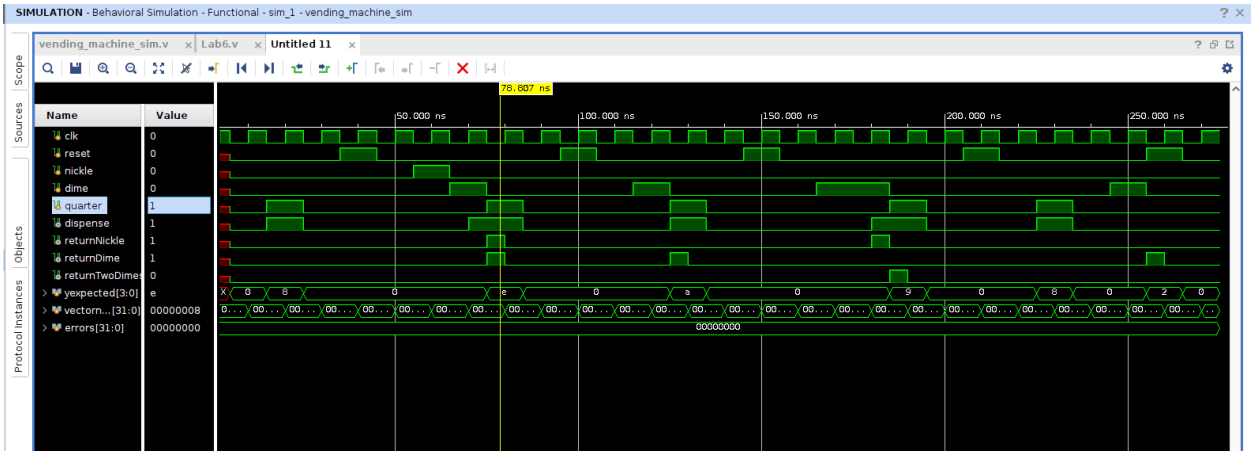
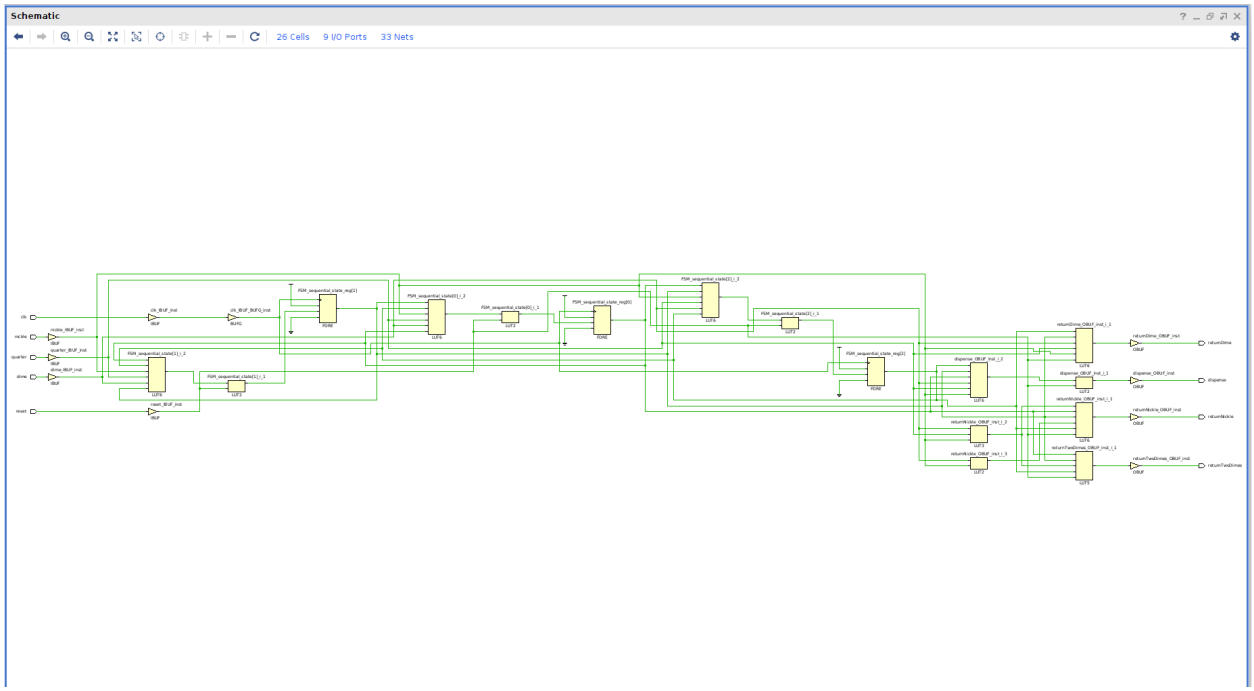
Alfredo De La Torre
CMPE 125 Lab - Digital Design 2
Section 04
11/5/25

CMPE 127
Vending Machine

Waveform Simulation:



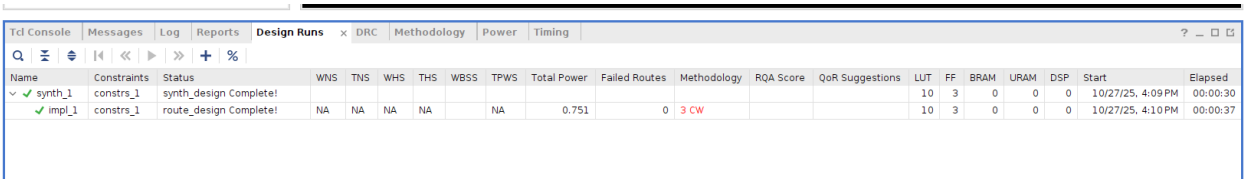
Schematic:



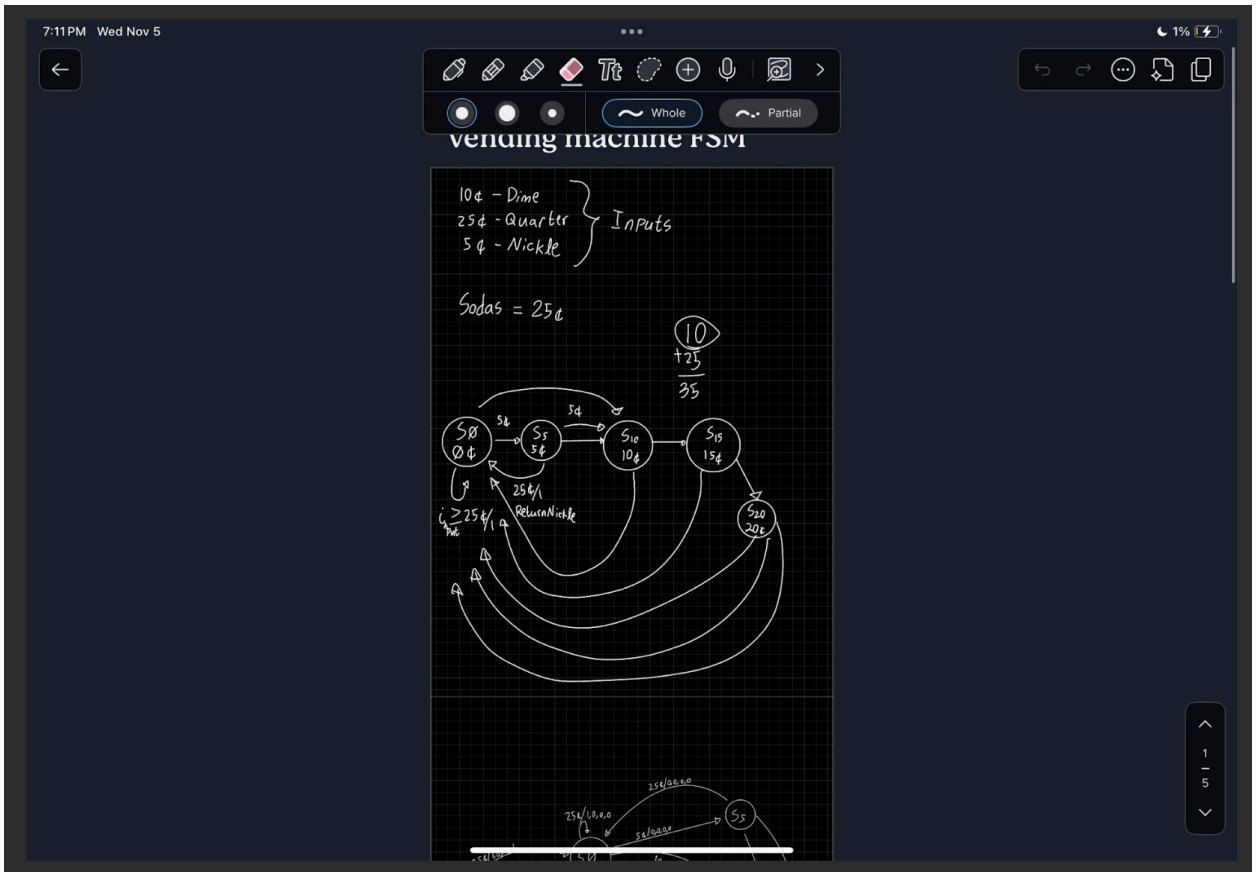
TCl Console:

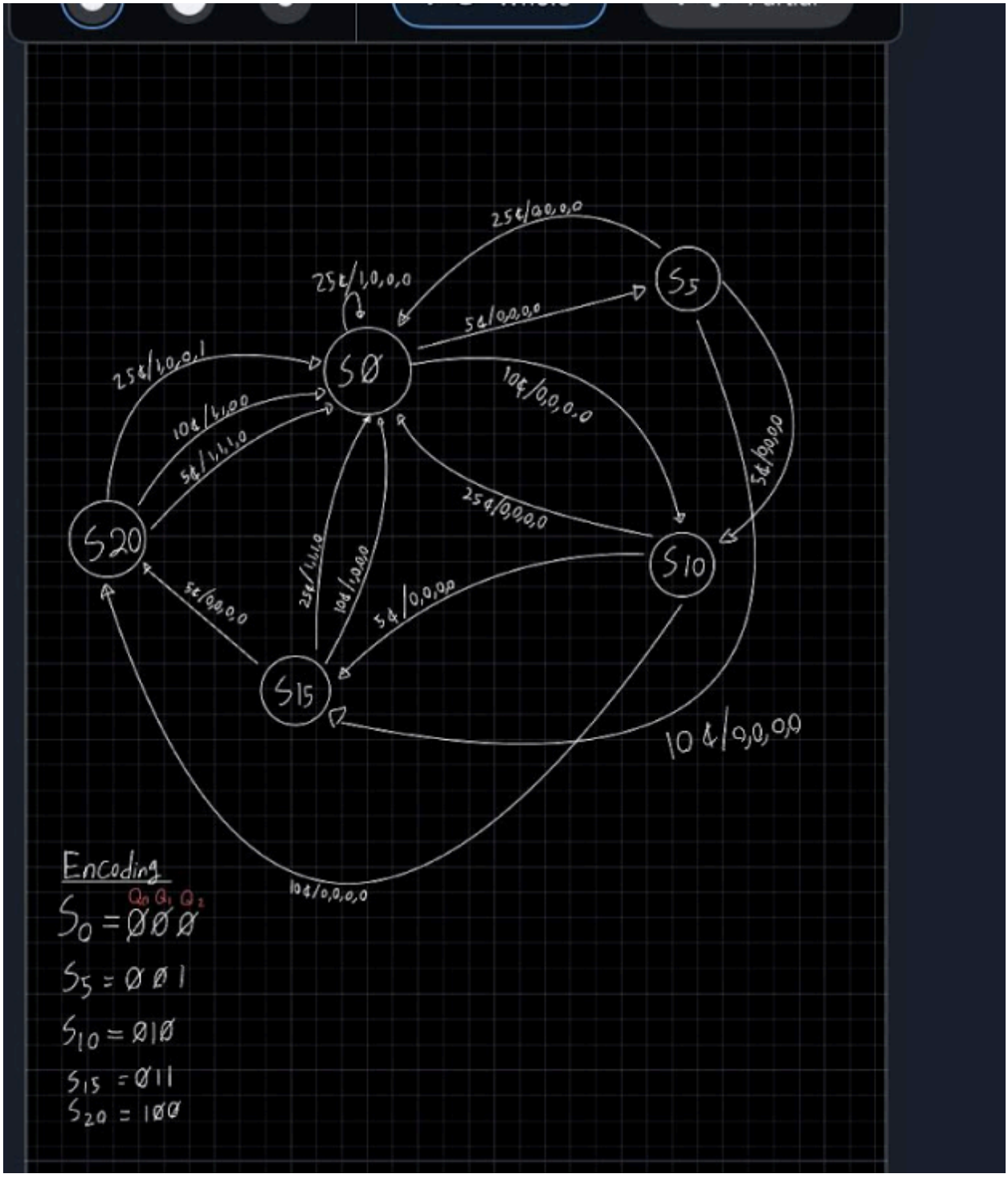
set curr_wave [current_wave_config]
if { [string length $curr_wave] == 0 } {
if { [llength [get_objects]] > 0} {
add_wave /
set_property needs_save false [current_wave_config]
} else {
send_msg_id Add_Wave-1 WARNING "No top level signals found. Simulator will start without a wave window. If you want to open a wave window go to 'File->New Waveform Configuration' or type 'create_wave_config' in the TCL console."
}
}
WARNING: [Wavedata 42-489] Can't add object "/vending_machine_sim/testvectors" to the wave window because it has 80008 bits, which exceeds the display limit of 65536 bits. To change the display limit, use the command "set_property display_limit <new limit> [current_wave_config]".
run 1000ns
28 tests completed with 0 errors

$finish called at time : 275 ns : File "/home/015634697@SJSUAD/CMPE127_Lab6/CMPE127_Lab6.srcs/sim_1/new/vending_machine_sim.v" Line 62
INFO: [USF-XSim-96] XSim completed. Design snapshot 'vending_machine_sim_behav' loaded.
INFO: [USF-XSim-97] XSim simulation ran for 1000ns
launch_simulation: Time (s): cpu = 00:00:05 ; elapsed = 00:00:05 . Memory (MB): peak = 8796.078 ; gain = 17.992 ; free physical = 1248 ; free virtual = 6208



PROOF:




25¢/0,0,0,0
25¢/1,0,0,0
S5
5¢/0,0,0,0
25¢/1,0,0,1
S0
10¢/0,0,0,0
10¢/1,0,0,0
5¢/1,1,1,0
5¢/0,0,0,0
25¢/0,0,0,0
S20
S10
25¢/1,1,1,0
10¢/1,0,0,0
5¢/0,0,0,0
5¢/0,0,0,0
S15
10¢/0,0,0,0
10¢/0,0,0,0
Encoding
Q0 Q1 Q2
S0 = 0 0 0
S5 = 0 0 1
S10 = 0 1 0
S15 = 0 1 1
S20 = 1 0 0

State Table

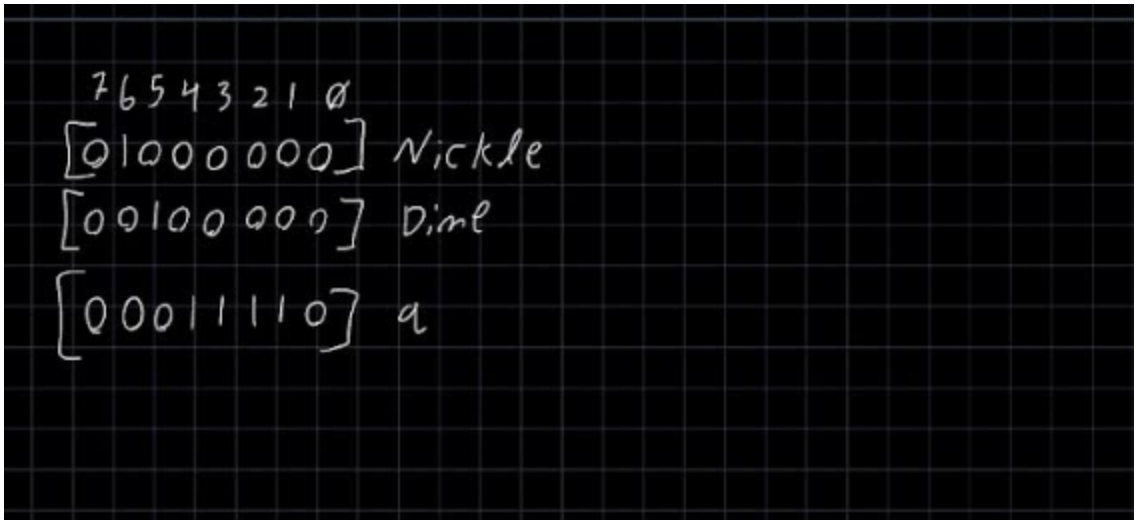
	Present State			Inputs			Next State			Outputs				D Flip-Flop		
	Q_0	Q_1	Q_2	N	D	Q	Q_0'	Q_1'	Q_2'	(D) Dispense	(RN) Return Nickle	(RD) Return Dime	(R2D) Return 2 Dime	D_0	D_1	D_2
0¢	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	1
	0	0	0	0	1	0	0	1	0	0	0	0	0	0	1	0
	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0
5¢	0	0	1	1	0	0	0	1	0	0	0	0	0	0	1	0
	0	0	1	0	1	0	0	1	1	0	0	0	0	0	1	1
	0	0	1	0	0	1	0	0	0	1	1	0	0	0	0	0
10¢	0	1	0	1	0	0	0	1	1	0	0	0	0	0	1	1
	0	1	0	0	1	0	1	0	0	0	0	0	0	1	0	0
	0	1	0	0	0	1	0	0	0	1	0	1	0	0	0	0
15¢	0	1	1	1	0	0	1	0	0	0	0	0	0	1	0	0
	0	1	1	0	1	0	0	0	0	1	0	0	0	0	0	0
	0	1	1	0	0	1	0	0	0	1	1	1	0	0	0	0
20¢	1	0	0	1	0	0	0	0	0	1	0	0	0	0	0	0
	1	0	0	0	1	0	0	0	0	1	1	0	0	0	0	0
	1	0	0	0	0	1	0	0	0	1	0	0	1	0	0	0

[reset, nickle, dime, quarter, dispense soda, return nickle, return Dime, return two Dimes]

[7, 6, 5, 4, 3, 2, 1, 0]

7 6 5 4 3 2 1 0

[00011000] quarter only, dispense Soda


7 6 5 4 3 2 1 0
[01000000] Nickle
[00100000] Dime
[00011110] a